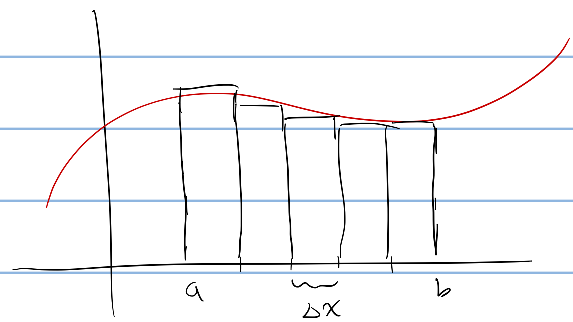
Recall Riemann Sum:

$$\sum_{k=1}^{n} f(x_k)\Delta x \approx \int_a^b f(x)\,dx \quad \text{and} \quad \Delta x = \frac{b-a}{n}$$

for n equal subintervals

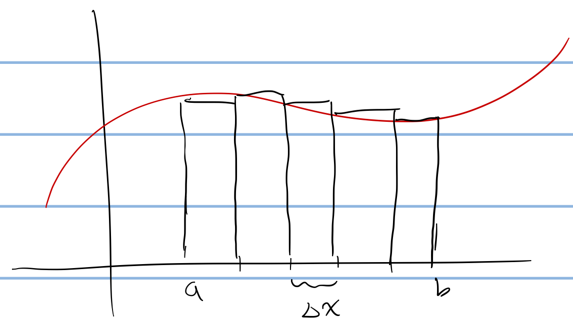
right endpoint

$x_k = a + k\Delta x$



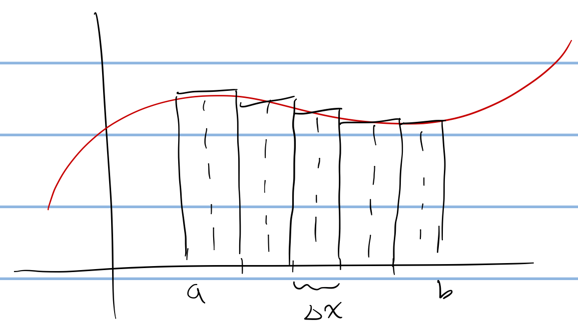
left endpoint

$x_k = a + (k-1)\Delta x$

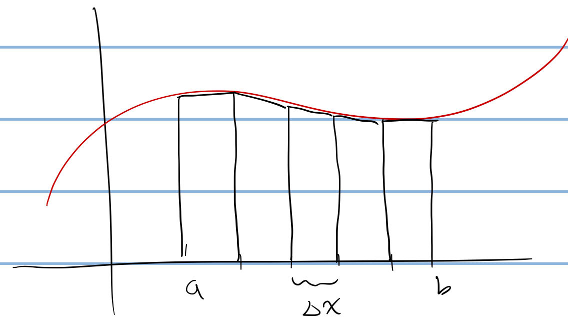


midpoint rule

$x_k = a + (k - \frac{1}{2})\Delta x$



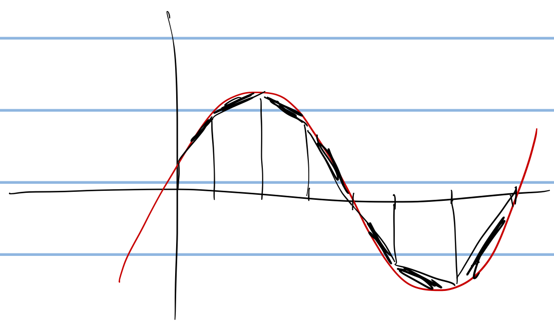
trapezoidal rule

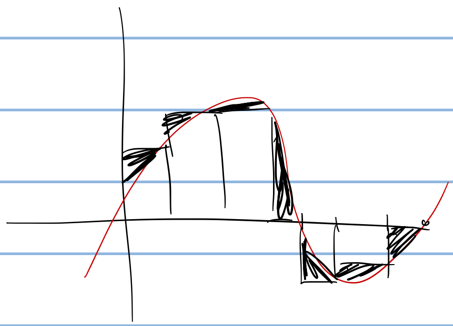


$$A = \frac{1}{2}b(h_1 + h_2)$$

$$\sum_{k=1}^{n} \frac{f(x_k) + f(x_{k-1})}{2} \Delta x$$

$$x_k = a + k\Delta x$$





$$\boxed{\sum_{k=1}^{n} f(x_k)\,\Delta x}$$

$$\Delta x = \frac{b-a}{n} = \frac{4}{n}$$

$$x_k = a + (k-1)\Delta x$$

$$= -1 + (k-1)\frac{4}{n}$$

$$\boxed{x_k^2 + 2x_k - 3}$$

$$\Delta x \sum \left(-1+(k-1)\frac{4}{n}\right)^2 + 2(-1+\cdots) - 3$$

$$\Delta x \quad \frac{3n^2 + 4n + 6}{5n^2}$$

$$\frac{16n^2 + \cdots + n}{3n^2}$$

$$\sum \left(-1+(k-1)\frac{4}{n}\right)^2$$

$$\Delta x \sum_{k=1}^{n} 1 - \frac{8}{n}(k-1) + (k-1)^2\frac{16}{n^2}$$

$$\frac{4}{n}\sum_{k=1}^{n} 1 \underbrace{- \frac{8}{n}k} + \frac{8}{n} + (k^2+2k-1)\frac{16}{n^2}$$

$$-\frac{8}{n}\sum_{k=1}^{n} k$$

$$\frac{-8}{n}\frac{n(n+1)}{2}$$

$$\underbrace{\sum_{k=1}^{n} \left(-1+\frac{4}{n}k\right)^2 + 2\left(-1+\frac{4}{n}k\right) - 3}$$

$$\Delta x = \frac{b-a}{n} = \frac{4}{n}$$

$$x_k = a + k\Delta x$$

$$= -1 + k\cdot\frac{4}{n}$$

$$\sum \frac{f(x_k)+f(x_{k-1})}{2}\Delta x$$

$$\frac{x_k^2 + 2x_k - 3 + x_{k-1}^2 + 2x_{k-1} - 3}{2}$$

$$\sum \frac{8k^2}{n^2} + \frac{3k}{n}$$

$$\frac{8}{n^2}\sum k^2$$

$$\frac{8}{n}\, n(n+1)(2n+1$$